

User Guide: Integrated Interpolation Suite

Group No: 3

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1 Introduction

Welcome to the Integrated Interpolation Suite! This is a command-line program designed to help you estimate values between known data points. Think of it like a smart "connect-the-dots" tool. If you have data from an experiment or a table, this program can calculate a reasonable guess for a value that you didn't measure.

The suite uses five different mathematical methods to give you a range of estimates. This guide will walk you through setting up, running the program, and understanding the different results it provides.

2 System Requirements

To compile and run this program, you will need:

- **A C++ Compiler:** The program is designed for Visual Studio 2010, so the **Microsoft C++ compiler (cl.exe)** is ideal.
- **A Plain Text Editor:** To create the source code file (e.g., Notepad, VS Code).

3 Setup and Execution

Follow these steps to get the program running.

Step 1: Create the Program File

Open your text editor, copy the entire C++ source code for the program, and save it in a new folder with the name `interpolation_suite.cpp`.

Step 2: Compile the Program

The best way to compile is by using the **Developer Command Prompt for VS2010**, which sets up all the necessary environment variables.

Open the Developer Command Prompt, navigate to the folder containing your file, and execute the following command:

```
cl /EHsc interpolation_suite.cpp
```

If the compilation is successful, a new executable file named `interpolation_suite.exe` will be created.

Step 3: Run the Program

In the same command prompt window, simply type the name of the executable and press Enter:

```
interpolation_suite.exe
```

4 How to Use the Program: A Walkthrough

The program will guide you through entering your data. Let's walk through a complete example.

Step 1: Enter the Number of Data Points

First, the program will ask how many data points you have. Let's say we have 3.

```
Enter number of data points (2 to 50): 3
```

Step 2: Enter the Data Points

Next, enter your data pairs, one per line, in the format `x y`. **Important:** You must enter them so that the x-values are sorted from smallest to largest.

```
Enter data points (x y) sorted by x:
Point 1: 1.0 2.0
Point 2: 3.0 5.0
Point 3: 6.0 7.0
```

Step 3: Enter the Target Value

Finally, enter the x-value for which you want to find an estimate. Let's find the value at $x = 4.0$.

```
Enter the target X value to interpolate: 4.0
```

5 Understanding the Output

After you provide all the inputs, the program will display a table of results.

----- CALCULATION RESULTS -----	
Target X: 4.0000	
Method	Result

1. Piecewise Constant	5.0000

2. Nearest Neighbor	5.0000
3. Linear Interpolation	5.6667
4. Newton Divided Diff	6.0000
5. Lagrange Polynomial	6.0000

Listing 1: Example program output for the walkthrough data.

Here is what each result means:

- **1. Piecewise Constant:** This method assumes the value is constant after a data point is passed. Since our target (4.0) is after the point (3.0, 5.0), it takes the value **5.0**. This creates a "step" pattern.
- **2. Nearest Neighbor:** This method simply finds the known data point closest to your target. Our target (4.0) is closer to $x=3$ than it is to $x=6$, so it returns the y -value from the point (3.0, 5.0), which is **5.0**.
- **3. Linear Interpolation:** This method draws a straight line between the two surrounding points (3.0, 5.0) and (6.0, 7.0) and finds the value on that line at $x=4.0$. This gives a more "averaged" estimate of **5.6667**.
- **4. Newton & 5. Lagrange:** These are advanced methods that create a single, smooth curve (a polynomial) that passes perfectly through *all* of your data points. They are the most mathematically complex and should always give the exact same result (**6.0000** in this case). They are best for modeling smooth, continuous phenomena.

6 Troubleshooting

If the program gives an error, refer to these common issues.

Error: Invalid input. Please enter an integer... You typed text or a decimal when the program was expecting a whole number (for the count of data points). Please restart and enter a valid integer.

Error: [CRITICAL ERROR]: Data is not sorted! You entered the data points out of order. For example, you entered (6.0, 7.0) before (3.0, 5.0). The program requires the x -values to be sorted from smallest to largest.

Error: [CRITICAL ERROR]: Duplicate X values detected! You entered two points with the exact same x -value, like (3.0, 5.0) and (3.0, 8.0). This is mathematically invalid for most interpolation methods. Please check your data.

Error: [Error: Target out of range...] You entered a target x -value that is smaller than your smallest x or larger than your largest x . Some methods (like Linear and Piecewise) cannot "extrapolate" and will show an error.